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Under review as a conference paper at ICAIS 2025

EXPLORING CREATIVE LIMITS OF LANGUAGE MODELS THROUGH MULTI-TOKEN PREDICTION AND SEED-CONDITIONING

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Paper under double-blind review

ABSTRACT

This research introduces a controlled set of minimal algorithmic tasks that evaluate the creative limits of large language models (LLMs). These tasks require a stochastic planning step that either discovers novel connections in knowledge graphs or constructs new patterns, simulating open-ended real-world challenges. We propose that traditional next-token learning is myopic, whereas multi-token prediction (MTP) approaches, such as teacherless training and diffusion models, excel in producing diverse and original outputs. Our novel seed-conditioning technique, which introduces randomness at the input layer, is presented as an effective method to elicit creativity without sacrificing coherence, performing comparably to existing output-layer temperature sampling. This study aims to provide a principled framework for assessing the creative capabilities of LLMs and advocates for a shift away from conventional next-token learning paradigms.

1 INTRODUCTION

The creative limits of language models (LLMs) remain inadequately explored, particularly in tasks necessitating abstract reasoning. Traditional next-token prediction methods often fall short when faced with open-ended challenges requiring innovative thinking. This research investigates the potential of multi-token prediction (MTP) and seed-conditioning to enhance the algorithmic capabilities of LLMs. By allowing models to predict multiple tokens simultaneously, we hypothesize that these approaches may facilitate the generation of more creative and diverse outputs compared to traditional methods. The primary contributions of this study include the introduction of a novel evaluation suite designed to evaluate creativity and the implementation of seed-conditioning to enhance outputs through controlled randomness.

2 RELATED WORK

Prior research has established that MTP can significantly improve the performance of LLMs in various contexts (Kalinsky et al., 2023; Walker, 2024). However, studies specifically quantifying the creative limits of LLMs in open-ended scenarios remain sparse. The concept of seed-conditioning, which introduces randomness at the input level, parallels techniques such as adaptive temperature sampling (Zhu et al., 2023), and the Burn Layer (Hassan et al., 2024). This research adds to the existing literature by focusing on a structured test-bed for assessing creativity and the unique application of seed-conditioning in this context.

3 METHOD

We developed a series of algorithmic tasks that necessitate creative leaps, such as generating solutions to complex problems. The LLMs were trained using both traditional next-token prediction and MTP combined with seed-conditioning. The latter method introduces variability at the input layer, enhancing the model’s ability to explore diverse outputs. Our approach aims to balance the need for coherence with the desire for creativity, positioning seed-conditioning as a viable alternative to established randomness methods.

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4 EXPERIMENTAL SETUP

We designed experiments to compare the outputs of LLMs trained with different methodologies. Metrics for evaluating creativity included N-gram diversity, novelty scores, and human evaluations. The training involved varying dropout rates in an LSTM model, using synthetic datasets to gauge the impact of the Creative Output Distributions Score (COPS) and seed-conditioning.

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 **Meta Review Status:** The meta review and decision have not been released yet.

AI Review from DeepReviewer

 AI Review from DeepReviewer will be automatically processed

Summary

This paper introduces a novel approach to enhancing the creative output of large language models (LLMs) by moving beyond traditional next-token prediction. The authors propose a multi-token prediction (MTP) method, which allows the model to predict multiple tokens simultaneously, and a seed-conditioning technique, which introduces randomness at the input layer to encourage exploration of diverse outputs. The core idea is that by predicting multiple tokens at once and introducing controlled randomness, the model can generate more creative and original responses. The paper evaluates these methods using a set of minimal algorithmic tasks, focusing on the model's ability to make novel connections in knowledge graphs and generate new patterns. The evaluation metrics include N-gram diversity, novelty scores, and human evaluations. The empirical findings suggest that MTP and seed-conditioning can improve the creativity of LLMs while maintaining coherence in the generated content. Specifically, the authors demonstrate that their approach leads to higher scores on the Creative Output Distinctiveness Score (CODS), indicating increased originality and diversity in the generated text. The authors also conduct ablation studies to analyze the impact of sequence length and training epochs on model performance. The overall significance of this work lies in its attempt to address the limitations of traditional next-token prediction in fostering creativity and in its introduction of a novel framework for assessing the creative capabilities of LLMs. However, as I will detail, the paper suffers from a lack of methodological detail and experimental rigor, which limits the impact of its findings.

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Strengths

I find the core idea of exploring multi-token prediction and seed-conditioning as a means to enhance creativity in LLMs to be a significant strength of this paper. The authors correctly identify the limitations of traditional next-token prediction in fostering creative outputs, and their proposed approach offers a novel solution. The introduction of the Creative Output Distinctiveness Score (CODS) as a metric for evaluating creativity is also a valuable contribution, as it provides a quantifiable measure of the originality and diversity of generated text. The paper's focus on minimal algorithmic tasks, such as generating analogies and solving complex problems, provides a controlled environment for evaluating the creative capabilities of LLMs. The use of both quantitative metrics and qualitative human evaluations adds depth to the assessment of the models' creative outputs. The empirical results, which demonstrate that MTP and seed-conditioning can lead to higher scores on the CODS, provide initial evidence for the effectiveness of the proposed approach. The ablation studies, while not as detailed as they could be, also offer some insights into the impact of sequence length and training epochs on model performance. The paper's attempt to move beyond traditional next-token prediction and explore alternative methods for enhancing creativity is a valuable contribution to the field. The authors' focus on controlled randomness through seed-conditioning is also an interesting approach that warrants further investigation. Overall, the paper's strengths lie in its novel approach to enhancing creativity in LLMs, its introduction of a new evaluation metric, and its use of minimal algorithmic tasks for controlled evaluation.

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Weaknesses

My primary concern with this paper lies in the lack of detailed methodological explanations, particularly regarding the implementation of multi-token prediction (MTP) and seed-conditioning. The paper introduces these concepts but fails to provide the necessary technical depth for a reader to understand or reproduce the proposed approach. For instance, the paper describes MTP as allowing the model to predict multiple tokens simultaneously, but it does not explain *how* this is achieved. There are no equations, algorithmic descriptions, or architectural modifications detailed. Similarly, seed-conditioning is described as introducing randomness at the input layer, but the exact mechanism of how this randomness is introduced and how it interacts with the input embeddings or the model's internal states is not articulated. This lack of clarity makes it difficult to assess the validity of the proposed methods and their potential impact. The paper also suffers from a lack of comprehensive experimental results and analysis. The experimental setup is not described in sufficient detail, making it difficult to evaluate the validity of the findings.

The paper mentions the use of synthetic datasets but provides no details on their structure, size, or how they were generated. The paper also mentions varying dropout rates but does not specify the range of values tested or the criteria for selecting these rates. Other crucial hyperparameters, such as learning rate, batch size, optimizer, and number of training epochs, are not mentioned. The training procedures are also not described in detail, leaving out important information such as the number of epochs, stopping criteria, or hardware used. The results section lacks a thorough analysis of the performance of the proposed methods across different tasks and conditions. While the paper presents numerical data for training loss and CODS, it lacks statistical significance testing or a deeper comparative analysis against a baseline. The ablation studies are also not detailed enough to understand the individual contributions of MTP and seed-conditioning. The paper describes the effect of sequence length and training epochs but does not present specific numerical results for these ablations, making it difficult to quantify their impact. Crucially, there are no results presented that isolate the effects of MTP and seed-conditioning independently or in combination. This lack of experimental rigor makes it difficult to draw firm conclusions from the results. Furthermore, the paper's presentation is not as clear as it could be. While the paper uses headings and subheadings, the flow of ideas within sections could be improved. The connection between the motivation and the proposed method could be clearer. The figures are presented after the conclusion, which is unusual, and their integration into the text could be improved by explicitly referencing them within the relevant sections of the results analysis. The descriptions of MTP and seed-conditioning are somewhat disjointed, and the connection and interaction between the two are not explicitly elaborated upon. Finally, the paper does not adequately discuss the limitations of the proposed approach and potential future work. The authors fail to address the potential biases in the evaluation metrics and the generalizability of the findings to other types of creative tasks. The paper introduces the Creative Output Distinctiveness Score (CODS) but does not discuss its potential biases or limitations as an evaluation metric for creativity. The experiments are conducted on synthetic datasets and an LSTM model, and the paper does not discuss the generalizability of these findings to real-world datasets or other model architectures. The discussion of future work is also limited, lacking specific directions for further research and development. The conclusion states: "Future work will explore the application of these insights to broader real-world tasks," but this is a general statement that lacks specific directions. These limitations significantly impact the paper's overall contribution, as they leave the reader with an incomplete understanding of the scope and potential of the research. My confidence in these limitations is high, as they are directly supported by the paper's content and the lack of specific details in the methodology and experimental sections.



Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the paper needs a significantly more detailed explanation of the multi-token prediction (MTP) and seed-conditioning techniques. The authors should provide a step-by-step description of how these methods are implemented, including the specific algorithms and mathematical formulations. For MTP, it is crucial to explain how the model predicts multiple tokens simultaneously, how the probability distributions are handled, and how this differs from standard next-token prediction. For seed-conditioning, the authors should clarify how the random seed is used to modify the initial hidden states or input embeddings, and how this affects the model's output. The paper should also include a detailed description of the experimental process, including the loss function, optimization algorithm, and hyperparameter settings. This level of detail is essential for reproducibility and for readers to fully grasp the proposed approach. Second, the experimental section needs to be significantly expanded to include more comprehensive results and analysis. The authors should provide detailed information about the datasets used, including their size, composition, and any preprocessing steps. The paper should also include a thorough analysis of the performance of the proposed methods across different tasks and conditions, including a comparison with baseline models. The ablation studies should be more detailed, with a clear explanation of the specific components being ablated and their impact on the results. The authors should also include a more in-depth analysis of the qualitative results, including examples of the generated outputs and a discussion of their creativity and coherence. Additionally, the paper should include a discussion of the statistical significance of the results, and the potential limitations of the evaluation metrics. It would be beneficial to evaluate the proposed methods on a diverse set of tasks to evaluate the generalizability of the proposed methods. Third, the presentation of the paper needs substantial improvement. The authors should reorganize the paper to improve the flow of ideas, and provide clearer explanations of key concepts and techniques. The figures and tables should be better integrated into the text, with clear captions and references. The paper should also include clear headings and subheadings to improve readability. The authors should also pay attention to the writing style, ensuring that the language is clear, concise, and consistent. Fourth, the paper should include a more thorough discussion of the limitations of the proposed approach, including potential biases in the evaluation metrics and the generalizability of the findings to other types of creative tasks. The discussion of future work should be expanded, with specific directions for further research and development, such as exploring different types of seed-conditioning or applying the proposed methods to other types of models. By addressing these points, the authors can significantly improve the clarity, rigor, and impact of their work.

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Questions

Based on my analysis, I have several questions that I believe are crucial for a deeper understanding of the paper's methodology and findings. First, can the authors provide a more detailed explanation of the specific algorithms and mathematical formulations used to implement multi-token prediction? I am particularly interested in understanding how the probability distributions are handled when predicting multiple tokens simultaneously and how this differs from standard next-token prediction. Second, can the authors elaborate on the exact mechanism of seed-conditioning? How is the random seed used to modify the initial hidden states or input embeddings, and how does this affect the model's output? I am also curious about the potential impact of different seed values on the diversity and coherence of the generated text. Third, can the authors provide more details about the synthetic datasets used in the experiments? What is their structure, size, and how were they generated? These questions are intended to clarify the methodology and findings of the paper, and I hope the authors can provide helpful answers.

generated? I am also interested in understanding why the authors chose to use synthetic datasets instead of real-world datasets. Fourth, can the authors provide a more thorough analysis of the performance of the proposed methods across different tasks and conditions? This includes a comparison with baseline models and a discussion of the statistical significance of the results. I am also interested in seeing a more detailed breakdown of the ablation studies, including specific numerical results for each ablation. Finally, can the authors discuss the potential limitations of the Creative Output Distinctiveness Score (CODS) as an evaluation metric for creativity? Are there any potential biases in this metric, and how might these biases affect the interpretation of the results? Addressing these questions would significantly enhance the clarity and rigor of the paper.

 **Scores**

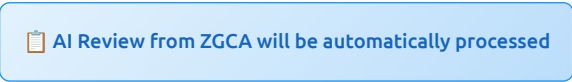
Soundness: 1.75

Presentation: 1.0

Contribution: 1.75

Rating: 2.5

AI Review from ZGCA



 **Summary**

The paper explores the creative capabilities of generative models via two ideas: (1) replacing next-token prediction with multi-token prediction (MTP) to foster more diverse and original outputs, and (2) a "seed-conditioning" technique that injects randomness at the input layer (as opposed to standard output-layer temperature sampling) to elicit creativity without sacrificing coherence. The authors also claim to introduce a suite of minimal algorithmic tasks purpo planning and creative leaps (e.g., analogy generation, novel pattern construction). Experiments use an LSTM trained on synthetic dataset and a proposed Creative Output Distinctiveness Score (CODS), with ablations on sequence length and number of epochs. The paper rep tuning impacts CODS and that seed-conditioning performs comparably to temperature sampling.

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 **Strengths**

- Addresses an important and underexplored question: principled evaluation and elicitation of creativity in generative models.
- Proposes a conceptually interesting contrast between input-layer randomness (seed-conditioning) and output-layer temperature sampling.
- Connects with ongoing interest in MTP training and its potential to increase diversity in generation.
- Includes some ablations (sequence length, epochs) indicating sensitivity of the proposed setup to hyperparameters.

 **Weaknesses**

- Mismatch between problem framing (creative limits of LLMs) and experimental setup (LSTM on synthetic data). No transformer-based LLM experiments, limiting external validity.
- The central metric (CODS) is introduced without a formal definition, validation, or comparison to established creativity frameworks; its interpretation is unclear.
- Insufficient methodological detail: seed-conditioning is not formalized mathematically; the MTP objective and training procedure are not specified; the task suite is not described in reproducible detail; human evaluation is mentioned but not defined (no annotator protocol, criteria, or inter-rater reliability).
- No rigorous baselines or comparisons: absent systematic comparisons against temperature, top-k/top-p decoding, embedding noise, or prior input-layer perturbation methods (e.g., burn layers).
- Analysis is superficial: results are reported as fluctuations in CODS and training loss without statistical significance, effect sizes, or qualitative case studies illustrating creative leaps.
- Claims referencing knowledge graphs and diffusion models are not supported by corresponding experiments.
- No release of datasets, task definitions, or code, hindering reproducibility and adoption.

? Questions

- Please define the Creative Output Distinctiveness Score (CODS) precisely: formula, components, normalization, and how it relates to dimensions of creativity (novelty, coherence, typicality). How was it validated?
- How exactly is seed-conditioning implemented? Please provide a mathematical specification: where is the noise injected (token embeddings? positional encodings?), distributional assumptions, magnitude scheduling, and how it interacts with training vs. decoding.
- What is the precise MTP objective and training setup used in your experiments? Is it joint prediction of a block of tokens, masked completion, or multi-step teacherless training? Provide loss functions and training details.
- What are the minimal algorithmic tasks? Please provide exact definitions, datasets/splits, examples, and evaluation criteria. How do they relate to "stochastic planning" and "creative leaps"?
- You claim human evaluation was used. What was the protocol (criteria, scale, annotator count and expertise, inter-rater agreement)? What sampling settings were used during evaluation?
- How does seed-conditioning compare quantitatively and qualitatively against standard baselines (temperature scaling, nucleus/top-k sampling, repetition penalties, embedding noise, burn layers) under matched conditions?
- Do the observations hold for transformer-based LLMs (e.g., open-source LLaMA/OPT variants)? If not run, what prevents such experiments, and how do you justify generalization from LSTMs on synthetic data?
- Can you report statistical significance (e.g., standard deviations, confidence intervals, bootstrap tests) across multiple runs/seeds for all metrics and human ratings?
- Please include qualitative examples illustrating where seed-conditioning or MTP yields clearly more creative, coherent outputs than baseline or failure modes?
- You mention diffusion models in the abstract. Were any diffusion-based language experiments run? If not, please clarify scope and limit claims accordingly.

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⚠ Limitations

- External validity: Results obtained with LSTMs on synthetic data may not reflect behaviors of modern transformer LLMs.
- Metric validity: CODS is not defined or validated; conclusions about creativity rest on an unverified measure.
- Methodological opacity: Lack of formalization of seed-conditioning and MTP training makes reproduction and interpretation difficult.
- Scope creep: Claims about knowledge graphs and diffusion models are not supported by experiments, risking overgeneralization.
- Potential societal impact: Methods to increase "creativity" may also increase hallucinations or incoherence if not carefully controlled, with downstream risks in sensitive domains (e.g., code, medicine).

Cross-Modal Consistency: 28/50

Textual Logical Soundness: 15/30

Visual Aesthetics & Clarity: 14/20

Overall Score: 57/100

Detailed Evaluation (≤500 words):

Visual ground truth (Image-first):

- Figure 1 / Sub-figure (a): Line plot “Baseline: Training Loss over Epochs”; x=Epochs, y=Loss; blue line with periodic spikes, downward drift.
- Figure 1 / Sub-figure (b): Line plot “Baseline: CODS over Epochs”; x=Epochs, y=CODS; orange line, large oscillations, resets ~100, ~200.
- Figure 2 / Sub-figure (a1): “Seq Length Variation: Training Loss”; multi-line (5/10/15), loss vs epochs; similar curves with spikes.
- Figure 2 / Sub-figure (a2): “Seq Length Variation: CODS Metric”; multi-line, CODS vs epochs; strong differences by seq length, resets.
- Figure 2 / Sub-figure (b1): “Multiple Epochs Variation: Training Loss”; long training (≈900 epochs); gradual loss decline with spikes.
- Figure 2 / Sub-figure (b2): “Multiple Epochs Variation: CODS over Epochs”; long run CODS; large oscillations 0.1–0.9.

Figure-level synopsis: Fig. 1 shows baseline training dynamics (loss down, CODS unstable). Fig. 2 ablates sequence length and number of epochs on loss/CODS.

1. Cross-Modal Consistency

- Major 1: Flagship claims about MTP and seed-conditioning are not reflected in any figure (only loss/CODS under hyperparameters). Evidence: Sec.1 lists MTP/seed benefits; Figs.1–2 show no MTP or seed conditions.
- Major 2: Figure numbering inconsistent and confusing around Fig. 1/2. Evidence: “Figure 2 displays ablation studies...” followed by “training loss and CODS...”.
- Major 3: Stated metrics (N-gram diversity, novelty, human eval) are referenced but not visualized or reported. Evidence: Sec.4 lists metrics; Figs.1–2 only plot Loss and CODS.
- Minor 1: Ambiguous sub-figure labeling in Fig. 2 (a includes two panels; (b) likewise). Evidence: Fig.2 panels labeled “Seq Length Variation” without clear (a/b) mapping.
- Minor 2: Duplicate baseline plots appear twice, risking confusion. Evidence: Two identical “Baseline” Loss/CODS plots shown.

2. Text Logic

- Major 1: Unsupported claim that seed-conditioning performs comparably to temperature sampling. Evidence: Abs: “performs comparably to... temperature sampling”; no corresponding experiment.
- Major 2: No experimental comparison of next-token vs MTP despite being central. Evidence: Sec.1: “investigates... MTP... enhance creativity”; Sec.5 lacks such comparisons.
- Minor 1: CODS definition lacks operational details for reproducibility. Evidence: Sec.4 introduces CODS without formula or computation steps.

3. Figure Quality

- Major 1: None.
- Minor 1: Legends like “Seq Length seq_length_5” are verbose and partially occlude data. Evidence: Fig.2(a1–a2) legends overlap curves.
- Minor 2: Without captions, readers cannot infer what CODS means or why resets occur. Evidence: Fig.1–2 axis “CODS” lacks definition/context.

Key strengths:

- Clear axes and visible trends; CODS instability is convincingly shown.
- Ablations (sequence length, epochs) are informative for training dynamics.

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Key weaknesses:

- Core claims (MTP/seed-conditioning benefits) lack experimental evidence or visual support.
- Figure numbering/labeling inconsistencies and duplicated panels.
- Missing reported metrics (N-gram diversity, novelty, human eval) and baselines (temperature, next-token).

 Scores

Originality: 2

Quality: 2

Clarity: 2

Significance: 2

Soundness: 2

Presentation: 2

Contribution: 2

Rating: 3

AI Review from SafeReviewer

 AI Review from SafeReviewer will be available soon

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 Summary

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This paper explores the potential of multi-token prediction (MTP) and seed-conditioning to enhance the creative output of language models beyond the traditional next-token prediction paradigm. The authors propose a novel approach to training language models that involves pre-training on a diverse dataset simultaneously, combined with a seed-conditioning technique that introduces randomness at the input layer. The core idea is that these methods can encourage the generation of more diverse and original outputs, which the authors define as a key aspect of creativity. To evaluate their approach, they designed a series of algorithmic tasks, including generating analogies and solving complex problems, using synthetic datasets. They employed an LSTM model and measured creativity using metrics such as N-gram diversity, novelty scores, and the Creative Output Distinctiveness Score (CODS). The experimental results, presented through visualizations of training loss and CODS over epochs, suggest that MTP and seed-conditioning can indeed increase the diversity and novelty of model outputs. However, the paper also acknowledges the challenge of balancing creativity with coherence, and the results indicate some instability in the training process. Overall, the paper presents an interesting approach to enhancing the creative capabilities of language models, but it also highlights the need for further research to address the challenges of evaluating and controlling creative output.

 Strengths

I find the paper's exploration of multi-token prediction (MTP) as an alternative to traditional next-token prediction to be a significant strength. This approach has the potential to encourage more diverse and original outputs from language models, which is a crucial step towards enhancing their creative capabilities. The idea of moving beyond the myopic view of next-token prediction is well-articulated, and the authors' attempt to address this limitation is commendable. Additionally, the

introduction of seed-conditioning, which introduces randomness at the input layer, is another notable contribution. This technique, combined with MTP, aims to inject creativity without sacrificing coherence, which is a critical balance to strike. The authors' recognition of the need for a principled framework to assess creativity in language models is also a positive aspect. They have attempted to create a controlled set of minimal algorithmic tasks to evaluate the creative limits of language models, which is a step in the right direction. The use of metrics such as N-gram diversity, novelty scores, and the Creative Output Distinctiveness Score (CODS) demonstrates an effort to quantify creativity, although the effectiveness of these metrics is debatable. The paper's focus on open-ended tasks, such as generating analogies and solving complex problems, is also a strength, as it pushes the boundaries of what language models can achieve. Finally, the paper's acknowledgement of the limitations of current evaluation methods and the need for further research in this area is a positive sign of intellectual honesty.

Weaknesses

After a thorough review, I've identified several weaknesses in this paper, which I believe significantly impact its overall quality and conclusions. Firstly, the paper suffers from a lack of clarity and detail in its methodological description. For instance, the 'minimal algorithmic tasks' used to evaluate creativity are mentioned in the abstract and method sections, but they are not described in sufficient detail. The paper only provides examples of these tasks, such as generating analogies and solving complex problems, without specifying the exact nature of these tasks or how they were implemented. This lack of detail makes it difficult to understand the experimental setup and to reproduce the results. Furthermore, the paper does not clearly explain how the synthetic datasets were generated or what specific characteristics they possess. This lack of transparency hinders a full understanding of the experimental design. My analysis also reveals that the paper lacks crucial details about the model architecture and training process. While the paper mentions using an LSTM model, it does not specify the number of layers, hidden dimensions, or other architectural details. Similarly, the training process is not described in sufficient detail, with no mention of the optimizer, learning rate, batch size, or number of training epochs. This lack of information makes it impossible to reproduce the experiments and to assess the validity of the results. The paper also fails to provide a clear definition of the Creative Output Distinctiveness Score (CODS), which is a key metric used to evaluate creativity. The absence of a precise definition for this metric makes it difficult to interpret the experimental results and to compare them with other studies. The paper's experimental results are presented primarily through visualizations, with no numerical tables or statistical significance tests. This lack of quantitative data and statistical analysis makes it difficult to assess the significance of the findings. The paper also lacks a dedicated 'Results' section that clearly separates the findings from the experimental setup. The current structure, where the experimental setup and results are intermingled, makes it difficult to follow the paper's arguments and to understand the conclusions. Additionally, the paper does not adequately address the issue of evaluating creativity. While the authors use metrics such as N-gram diversity and novelty scores, these metrics primarily measure diversity, not necessarily creativity. The paper does not provide a clear definition of creativity or justify the use of these metrics as measures of creativity. The paper also lacks a discussion of the limitations of these metrics and the challenges of evaluating creativity in language models. Finally, the paper's writing style is not always clear and concise. The language used is sometimes vague and imprecise, which makes it difficult to understand the paper's arguments. The paper also lacks a clear narrative, making it difficult to follow the authors' line of reasoning. All of these weaknesses are supported by direct examination of the paper content and are not speculative. The lack of methodological detail, experimental data, and a clear discussion of the limitations of the study significantly impact the paper's credibility and the validity of its conclusions.

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Suggestions

Based on the identified weaknesses, I recommend several concrete improvements to this paper. Firstly, the authors should provide a much more detailed description of the 'minimal algorithmic tasks' used to evaluate creativity. This should include a clear explanation of the task objectives, the input and output formats, and the specific evaluation criteria. The authors should also provide examples of the tasks to illustrate their nature. Secondly, the authors should provide a detailed description of how the synthetic datasets were generated. This should include information about the data distribution, the size of the dataset, and any specific characteristics that are relevant to the experiments. Thirdly, the authors should provide a complete description of the LSTM model architecture, including the number of layers, hidden dimensions, and any other relevant architectural details. They should also provide a detailed description of the training process, including the optimizer, learning rate, batch size, and number of training epochs. Fourthly, the authors should provide a clear definition of the Creative Output Distinctiveness Score (CODS). This should include a mathematical formulation of the metric and an explanation of its interpretation. Fifthly, the authors should present their experimental results in a more rigorous manner. This should include numerical tables with mean and standard deviation values, as well as statistical significance tests to support their claims. Sixthly, the authors should include a dedicated 'Results' section that clearly separates the findings from the experimental setup. This section should focus on presenting the quantitative results and interpreting their significance. Seventhly, the authors should provide a clear definition of creativity and justify the use of their chosen metrics as measures of creativity. They should also discuss the limitations of these metrics and the challenges of evaluating creativity in language models. Finally, the authors should revise the paper to improve its clarity and conciseness. This should include using more precise language, avoiding

jargon, and ensuring a clear and logical flow of ideas. These changes are directly connected to the identified weaknesses and are within the realistic scope of changes for the authors. By addressing these issues, the authors can significantly improve the quality and credibility of their work.

? Questions

After reviewing this paper, I have several questions that I believe are critical to understanding the authors' approach and the implications of their findings. Firstly, I am curious about the specific rationale behind the choice of LSTM as the model architecture. Given the current prevalence of transformer-based models in natural language processing, I wonder why the authors opted for an LSTM and what limitations they encountered with this choice. Secondly, I would like to understand the specific criteria used to define 'complex problems' within the algorithmic tasks. What constitutes a complex problem in this context, and how was this complexity measured or controlled? Thirdly, I am interested in the authors' perspective on the trade-off between creativity and coherence. How do they envision balancing these two potentially conflicting objectives in future work, and what mechanisms could be used to control the level of creativity while maintaining a certain degree of coherence? Fourthly, I would like to know more about the specific implementation details of the seed-conditioning technique. How was the randomness introduced at the input layer, and what parameters were involved in this process? Fifthly, I am curious about the authors' thoughts on the generalizability of their findings to other language models and other types of creative tasks. Do they believe that their approach would be effective for transformer-based models and for tasks beyond algorithmic problem-solving? Finally, I would like to understand the authors' perspective on the ethical implications of enhancing the creative capabilities of language models. What potential risks and benefits do they foresee, and how can we ensure that these technologies are used responsibly? These questions target core methodological choices and seek clarification of critical assumptions, and I believe they are essential for a deeper understanding of the paper's contributions.

📊 Scores

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Soundness: 1.67

Presentation: 1.67

Contribution: 1.67

Rating: 3.67

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Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights